Min June Yang

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Research Interests

Reinforcement learning and robot learning for mobile robots in shipbuilding and shipyard environments; robot path and motion planning; trajectory optimization; imitation learning; reinforcement learning for complex robotic tasks such as traversing stiffeners, manipulation, and coordinated mobile-manipulator behaviors.

Education

Seoul National University Mar. 2020 – Present
B.S. in Naval Architecture and Ocean Engineering; Interdisciplinary Major in Artificial Intelligence *Expected Feb. 2027*

- Academic focus: robotics, machine learning, reinforcement learning, control, signal processing, probability, linear algebra, and computing for maritime engineering applications.

Research Experience

KAIST DRCD Hubo Lab / DRCD Lab Jan. 2026 – Feb. 2026
Research Intern – Extended Lagrangian AGHF for Manipulator Trajectory Optimization *Daejeon, Korea*

- Participated in EL-AGHF research that reformulates AGHF-based trajectory optimization as an Extended Lagrangian / Lagrangian-dual problem to suppress motion in inadmissible control directions without relying only on excessively large penalty coefficients.
- Developed a MATLAB visualizer and simulator for validation on a planar 3-link, 3-joint manipulator, including forward kinematics, trajectory animation, final-state plotting, and comparison among AGHF solution trajectories, full-control integrated paths, and admissible-control integrated paths.
- Implemented the control-affine system model for the manipulator, including inertia, Coriolis, gravity, drift dynamics, admissible / inadmissible direction matrices, Riemannian metric construction, and kinematic joint-limit terms.
- Built an initial MATLAB solver prototype, then contributed to a faster C++ trajectory-optimization solver using CasADi symbolic differentiation to construct the extended Lagrangian, compute Euler-Lagrange PDE terms, solve the resulting DAE/ODE system, extract admissible controls, and integrate trajectories.
- Supported AGHF vs. EL-AGHF comparison experiments under different penalty coefficients and joint-limit settings; the implemented C++ pipeline reproduced low terminal-error EL-AGHF behavior while reducing solving time relative to the initial MATLAB implementation.

Selected Projects

MERO Robotics Club, Seoul National University Nov. 2025 – Dec. 2025
AGHF-Based Model-Based Control for a 2-Link Manipulator *Seoul, Korea*

- Built a MATLAB planner implementing an Affine Geometric Heat Flow (AGHF) / PHLAME-inspired trajectory optimization pipeline for a 2-DoF planar manipulator, interfacing with a Simulink robot-arm simulation and tracker.
- Formulated 2-link manipulator dynamics, including inertia, Coriolis/gravity, and friction terms, and generated joint-space trajectories from workspace targets using inverse kinematics.
- Implemented pseudospectral discretization with Chebyshev nodes and a differentiation matrix, converted the AGHF evolution into an ODE solve with `ode45`, and exported `q_star`, `dq_star`, and `u_star` timeseries for Simulink execution.
- Added a feedforward torque plus PD feedback tracking controller in Simulink, enabling the end-effector to converge to a target workspace coordinate in simulation.
- Documented the theoretical background, usage workflow, parameter tuning procedure, validation result, and limitations for future extension to constraints such as obstacle avoidance and joint limits.

How to Make a Robot with Artificial Intelligence, SNU Fall 2025
ROS2 Mobile-Robot Navigation Project: Localization-Aware Path Planning *Seoul, Korea*

- Contributed to a team project for a Unitree Go1 robot in Gazebo/ROS2, integrating localization, path planning, and perception modules for language-commanded navigation missions.
- Took responsibility for the path-planning component that consumes estimated robot pose from a particle-filter localization node and publishes navigation paths to an MPPI-based path tracking controller.

- Extended the provided A* / path-planning pipeline for map-based collision avoidance by treating perceived 2D obstacles such as walls with an additional safety buffer, effectively inflating occupied regions before planning.
- Integrated launch files and ROS2 topics so the planner, simulator, and path tracker could run together; the tracking stack used `/local_path` as the desired path, `/go1_pose` as feedback, and `/cmd_vel` for robot velocity commands.
- Gained practical experience with ROS2 package integration, occupancy-grid maps, localization-planning interfaces, MPPI tracking, and perception-aware autonomous navigation under imperfect localization and sensing.

Selected Coursework and Research-Relevant Training

- **Introduction to Robotics** – Rigid body motions, homogeneous transforms, forward/inverse kinematics, Jacobians, dynamics, path/trajectory planning, control, sensors/actuators, and computer vision.
- **How to Make a Robot with Artificial Intelligence** – Hands-on intelligent robotics using ROS; Bayesian estimation, particle filters, perception, mapping, localization, decision making, planning, control, VLA, and team project work.
- **Deep Learning (in progress, Spring 2026)** – ML foundations, MLE/MAP, bias-variance and inductive bias, MLP/CNN/RNN/LSTM/Seq2Seq, optimization, Transformers, VAE/GAN/diffusion, continual learning, LLM fine-tuning, VLM/VLA/world models, reinforcement learning basics, PPO, and GRPO.
- **Machine Learning for Naval Architecture and Ocean Engineering** – Statistical inference, probability, linear algebra, optimization, MLE, unsupervised learning, classification, linear regression, HMM, Bayesian inference, CNN/RNN, reinforcement learning, and machine-learning projects for naval architecture and ocean engineering applications.
- **Foundations of Machine Learning** – Generalization, bias-variance, overfitting, classification, linear/logistic regression, neural networks, kernel methods, SVM, probabilistic ML, Bayesian networks, naive Bayes, EM/GMM, HMM, CNN/RNN, ensemble learning, Python programming assignments, and final programming project.
- **Signal Processing / Signals and Systems** – Continuous/discrete-time signals and systems, Fourier series/transform, DTFT, time/frequency characterization, sampling, Laplace and Z transforms, linear feedback systems, MATLAB/Simulink exercises, and marine-engineering signal processing applications.
- **Mathematical and Computing Foundations** – Linear Algebra (Strang-based), Probability Variables and Stochastic Processes (Yates-based, in progress), Data Structures, Computer Programming in Java/C++, and model-based 2D/3D Computer Vision (in progress).

Technical Skills

Programming	Python, C++, MATLAB, Java
Robotics / Simulation	ROS2, Gazebo, MATLAB/Simulink; manipulator simulation, model-based control, and trajectory-generation experience
Core Methods	Motion planning, trajectory optimization, kinematics/dynamics, Bayesian estimation and filtering, machine learning, deep learning, reinforcement learning fundamentals
Languages	Korean (native), English (technical coursework and documentation)

Scholarships and Honors

- Gwacheon City Ae-Hyang Scholarship — May 2025, Sep. 2025
- SNU College of Engineering Scholarship — Oct. 2024
- SNU Internal Scholarship for Academic Excellence — Feb. 2025
- SNU Department of Naval Architecture and Ocean Engineering Jung-Gong Scholarship — Feb. 2024, Jun. 2025

Target Research Fit

My long-term goal is to develop robot learning and planning methods that enable mobile robots and mobile manipulators to operate robustly in shipbuilding environments. I am particularly interested in combining model-based planning, trajectory optimization, imitation learning, and reinforcement learning to address unstructured industrial conditions such as narrow passages, stiffener traversal, localization/perception uncertainty, and manipulation tasks in shipyards.